

PORTE LOGICHE → OPERANO SU **SEGNALI BINARI**

LIVELLI DI TENSIONE →
ALTO (1 / TRUE)
BASSO (0 / FALSE)

VARIABILI BOOLEANE → VARIABILE **BINARIA** CHE PUÒ ASSUMERE 1 o 0.

→ $X=1, Y=1, Z=0$

OPERATORI BOOLEANI → **NOT, AND E OR**
NOT → **NEGAZIONE LOGICA**
AND → **SOMMA LOGICA**
OR → **SI POSSONO CREARE ESPRESSIONI BOOLEANE.**

NOT → NOT X OPPURE $\bar{X}$ | **AND** → X AND Y OPPURE $X \cdot Y$ | **OR** → X OR Y OPP. $X + Y$

X	$\bar{X}$
1	0
0	1

X	Y	X AND Y
0	0	0
0	1	0
1	0	0
1	1	1

X	Y	X OR Y
0	0	0
0	1	1
1	0	1
1	1	1

POSSIAMO ESPRIMERE UNA **FUNZIONE BOOLEANA** UTILIZZANDO LA **TAVOLA DELLA VERITÀ**:

ES:

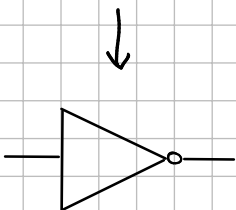
$$\bar{X}Y + X\bar{Y}$$

X	Y	$\bar{X}Y$	$X\bar{Y}$	f
0	0	0	0	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

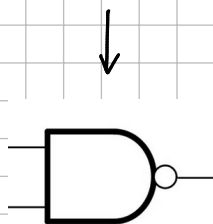
⇒ **XOR**

PORTE LOGICHE → ESPRIMONO UNA **FUNZIONE BOOLEANA**, HANNO **N** INGRESSI E 1 USCITA.

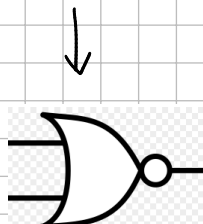
NOT



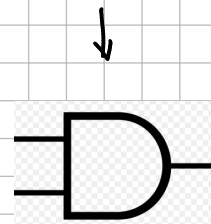
NAND



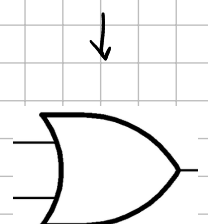
NOR



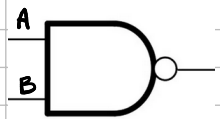
AND



OR



UNA PORTA LOGICA $\rightarrow$ INGRESSO $\rightarrow$ SEGNALI BINARI. (OPERANDI).
 $(0, 1)$
 $\rightarrow$ USCITA $\rightarrow$ RISULTATO.



$\Rightarrow$ LA NAND QUANDO RICEVE 2 SEGNALI ALTI CIOE' : $A=1$ E $B=1$
 DA COME RISULTA UN SEGNALE BASSO.

$\rightarrow$ $\boxed{\overline{A \cdot B}}$

X	Y	X NAND Y
0	0	1
0	1	1
1	0	1
1	1	0

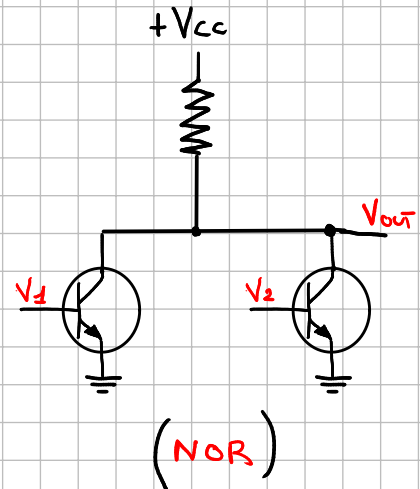
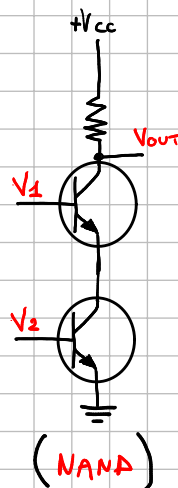
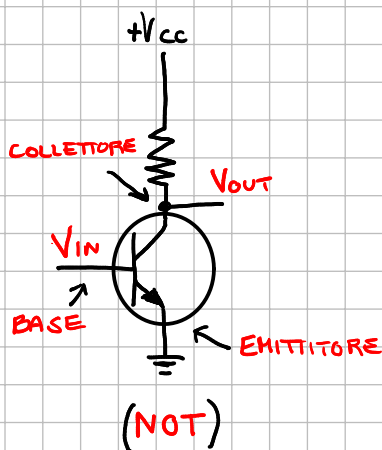
$\Rightarrow$ LA NEGAZIONE DI UNA
AND.

TRANSISTOR $\rightarrow$ SI COMPORTANO DA INTERRUPTORI $\rightarrow$ 3 CONNESSIONI $\rightarrow$ BASE $\rightarrow$ EMITTITORE

$\rightarrow$ QUANDO V_{IN} NELLA BASE E' BASSO, L'INTERRUPTORE E' APERTO E IL COLLETTORE V_{OUT} E' ALTO (NON SI SCARICA)

$\rightarrow$ COLLETTORE

$\rightarrow$ QUANDO V_{IN} E' ALTO, L'INTERRUPTORE E' CHIUSO E IL COLLETTORE V_{OUT} SI SCARICA ALLA BASE.



PROPRIETA' DEGLI OPERATORI LOGICI:

- NOT $\rightarrow \overline{\overline{X}} = X$
- AND $\rightarrow X \cdot 0 = 0; X \cdot 1 = X; X \cdot X = X; X \cdot \overline{X} = 0;$
- OR $\rightarrow X + 0 = X; X + 1 = 1; X + X = X; X + \overline{X} = 1;$
- NAND $\rightarrow \overline{X \cdot 0} = 1; \overline{X \cdot 1} = \overline{X}; \overline{X \cdot X} = \overline{X}; \overline{X \cdot \overline{X}} = 1;$
- NOR $\rightarrow \overline{X + 0} = \overline{X}; \overline{X + 1} = 0; \overline{X + X} = \overline{X}; \overline{X + \overline{X}} = 0;$

ALTRE PROPRIETA' :

$X(Y + 1) = X$; $X(Y + \overline{Y}) = X$ $\Rightarrow$ ANCHE SOSTITUENDO $\overline{X}$ A X .

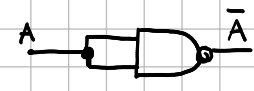
$X(Y + Z) = XY + XZ$; $X + YZ = (X + Y)(X + Z) = \overbrace{X \cdot X}^X + XZ + XY + YZ = X(1 + Z + Y) + YZ = X + YZ$

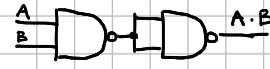
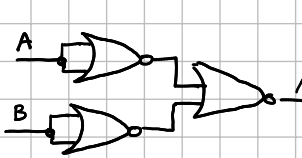
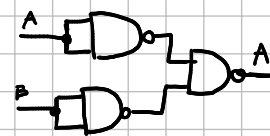
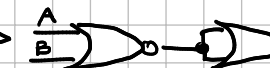
TEOREMA ASSORBIMENTO:

$$X + XY = X(1 + Y) = X; \quad X(X + Y) = X \cdot X + XY = X + XY = X \cdot (1 + Y) = X$$
$$(X + \bar{Y}) \cdot Y = XY + \bar{Y} \cdot Y = XY; \quad X\bar{Y} + Y = Y + X\bar{Y} = (Y + X)(Y + \bar{Y}) = X + Y$$

TEOREMI DI DE MORGAN:

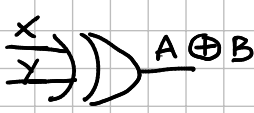
$$\overline{X \cdot Y} = \bar{X} + \bar{Y} \quad \overline{X + Y} = \bar{X} \cdot \bar{Y}$$

NAND → OPERATORE UNIVERSALE ⇒ **NOT** →  $\bar{A} \cdot \bar{A} = \bar{A}$
NOR →  $\bar{A} + \bar{A} = \bar{A}$

AND →  $A \cdot B$ $\overline{A \cdot B} = \bar{A} + \bar{B}$
→  $A \cdot B$ $\overline{A + B} = \bar{A} \cdot \bar{B}$
OR →  $A + B$ $\overline{A + B} = \bar{A} \cdot \bar{B}$
→  $A + B$ $\overline{\bar{A} \cdot \bar{B}} = A + B$

XOR → OPERATORE ESCLUSIVO

$$\bar{X}Y + X\bar{Y}$$



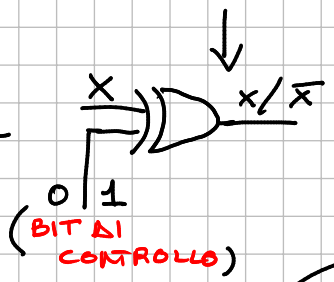
X	Y	X XOR Y
0	0	0
0	1	1
1	0	1
1	1	0

QUANDO SONO DIVERSI VALE 1.

$X \text{ XOR } 0 = X; \quad X \text{ XOR } 1 = \bar{X}; \quad X \text{ XOR } X = 0 \quad X \text{ XOR } \bar{X} = 1; \quad X \text{ XOR } Y = \bar{X} \text{ XOR } \bar{Y}$

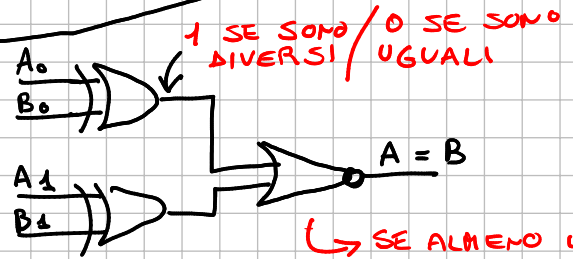
POSSIAMO REALIZZARE → INVERTITORE CONTROLLATO, COMPARATORE, VERIFICATORE DI PARITÀ.

SE L'INPUT È 0 IN USCITA ABBIAMO X SENNO $\bar{X}$



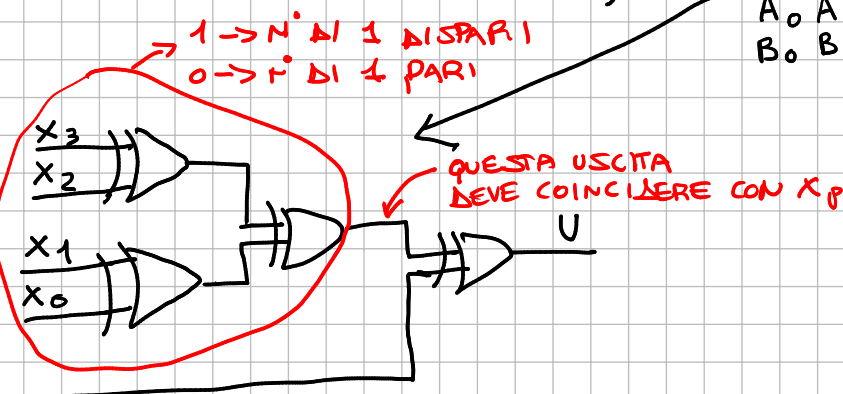
INPUT: 2 SEQUENZE DI 2 BIT

A₀ A₁
B₀ B₁



1 SE SONO DIVERSI / 0 SE SONO UGUALI

SE ALMENO UNO È MEZZO A 1 ALLO IL NOR DA 0.



QUESTA USCITA DEVE COINCIDERE CON X_P

X_P → 1 SE I BIT A 1 SONO DISPARI, SENNO 0.